

The limit

$$\lim_{x \rightarrow \infty} x^2 - x.$$

is

Select the correct option



does not exist.

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2.



0.



6.

The $\lim_{x \rightarrow 2} (3x - 5) =$

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Select the correct option



0.



1.



-1.



-2.

For the function $\frac{|x|}{x}$, identify the false statement(s)

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Select the correct option



$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$$



$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1.$$



$$\lim_{x \rightarrow 0} \frac{|x|}{x} \text{ doesn't exist.}$$



$$\lim_{x \rightarrow 0} \frac{|x|}{x} = 1.$$

For the function $f(x) = x \sin \frac{1}{x}$, $x \neq 0$, which statement is true

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Select the correct option



$$\lim_{x \rightarrow 1/\pi} f(x) = 0.$$



The function is not defined at

$$x = 0.$$



$$\lim_{x \rightarrow 0} f(x) = 1.$$



$$\lim_{x \rightarrow 0} f(x) = 0.$$

For the function defined as

$$f(x) = cx,$$

for every $\epsilon > 0$ the formal definition ensures

Select the correct option



$$|f(x) - cx_0| < \epsilon, \quad |x - x_0| < \delta, \text{ where } 0 < \delta < \epsilon.$$

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$$|f(x) - cx_0| < \epsilon, \quad |x - x_0| < \delta, \text{ where } 0 < \delta < \epsilon/|c|.$$



$$|f(x) - cx_0| < \epsilon, \quad |x - x_0| < \delta, \text{ where } -\epsilon < \delta < \epsilon.$$



$$|f(x) - cx_0| < \epsilon, \quad |x - x_0| < \delta, \text{ where } 0 < \delta < \epsilon.$$

The value of $x + \lim_{x \rightarrow 0+} \frac{|x|}{x} =$

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Select the correct option



$$x - 1.$$



$$x + 1.$$



$$x.$$



$$x + 1/x.$$

The limit

$$\lim_{x \rightarrow 1} \sqrt{x} - 1.$$

Select the correct option



1.

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0.



does not exist.



-1.

If $f(x) = \log x$ and $g(x) = \frac{1}{1-x^2}$, then $f \circ g$ is

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Select the correct option



$$(f \circ g)(x) = \log \frac{1}{1-x^2}.$$



$$(f \circ g)(x) = \log \frac{x}{1-x^2}.$$



$$(f \circ g)(x) = \log \frac{1}{x^2}.$$



$$(f \circ g)(x) = \log \frac{-1}{1-x^2}.$$

The $\lim_{x \rightarrow 0} \frac{4 - 4 \cos x - 2 \sin^2 x}{x^4}$ is

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Select the correct option



$\frac{1}{4}.$



$-\frac{1}{2}.$



$\frac{1}{2}.$



$2.$

If $\lim_{x \rightarrow 0} f(x) = 10$ and $\lim_{x \rightarrow 0} g(x) = -2$ then $\lim_{x \rightarrow 0} \left(\frac{f(x)}{g(x)} \right)$ is

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Select the correct option



-10.



5.



0.



-5.

For the function defined as $f(x) = cx$, for every $\epsilon > 0$ the formal definition of limit of function ensures

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Select the correct option

☐	$ f(x) - cx_0 < \epsilon, \quad x - x_0 < \delta, \text{ where } 0 < \delta < \epsilon.$
☒	$ f(x) - cx_0 < \epsilon, \quad x - x_0 < \delta, \text{ where } 0 < \delta < \epsilon/ c .$
☐	$ f(x) - cx_0 < \epsilon, \quad x - x_0 < \delta, \text{ where } -\epsilon < \delta < \epsilon.$
☐	$ f(x) - c < \epsilon, \quad x - x_0 < \delta, \text{ where } 0 < \delta < \epsilon.$

The value of the limit

$$\lim_{x \rightarrow 0^+} x \log x,$$

Select the correct option



0

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1



2



3

For the piecewise defined function $f(x) = \begin{cases} \end{cases}$

$$\begin{cases} x^3, & x \leq 0, \\ x^2 \sin \frac{1}{x}, & x > 0, \end{cases}$$

Select the correct option



0

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2



1



-1

The value of $x + \lim_{x \rightarrow 0^+} \frac{|x|}{x} =$

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Select the correct option



$$x - 1.$$



$$x + 1.$$



$$x.$$



$$x + 1/x.$$

Rolle's Theorem says, suppose that f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) and $f(a) = f(b)$.

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Select the correct option

☒	Then $f'(c) = 0$ for some c in the open interval (a, b) .
☐	Then $f'(c) \neq 0$ for some c in the open interval (a, b) .
☐	Then $f'(c)$ exists for some c in the open interval (a, b) .
☐	Then $f'(c) = 0$ for some c in the open interval (a, ∞) .

The function $g(x) = x^2$ is

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Select the correct option



decreasing on $[0, \infty)$.



increasing on $[0, \infty)$.



having negative values on $[0, \infty)$.



undefined on $[0, \infty)$.

The derivative of the function x^n is

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Select the correct option



$x^{n-1}.$



$nx^{n-1}.$



$\frac{n}{x^{n-1}}.$



$nx^n.$

The function $h(x) = -x^3$ is

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Select the correct option



is undefined on $(-\infty, \infty)$.



decreasing on $(-\infty, \infty)$.



increasing on $(-\infty, \infty)$.



positive on $(-\infty, \infty)$.

If a function f is continuous on the closed interval then f attains its

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Select the correct option



extreme values in the closed interval.



extreme values at one point in the closed interval.



may or may not attain extreme values in the closed interval.



derivative in the closed interval.

The Mean Value Theorem says, suppose that f is differentiable on $[a, b]$, $f'(a) \neq f'(b)$, and μ is between $f'(a)$ and $f'(b)$.

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Select the correct option



Then $f'(c) \neq \mu$ for some c in (a, b) .



Then $f'(c) = \mu$ for some c in (a, b) .



Then $f'(c) = 0$ for some c in (a, b) .



Then $f'(c)$ may be 0 or μ for some c in (a, b) .

The inverse of the function $f(x) = x^2$, is

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Select the correct option



$$f^{-1}(y) = y^2$$



$$f^{-1}(y) = \sqrt{y}.$$



$$f^{-1}(y) = \frac{1}{x^2}$$



$$f^{-1}(y) = \frac{1}{y^2}$$

The inverse of the given function $f(x) = 2x + 4, 0 \leq x \leq 2$, is

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Select the correct option



$$f^{-1}(y) = \frac{y - 4}{2}$$



$$f^{-1}(y) = \frac{y - 2}{4}$$



$$f^{-1}(y) = \frac{x - 2}{4}$$



$$f^{-1}(y) = \frac{y - 4}{4}$$

Generalized Mean Value Theorem says, If f and g are continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , then

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Select the correct option



for some c in (a, b) .

$$[g(b) - g(a)]f'(c) = [f(b) - f(a)]g'(c)$$



for some c in (a, b) .

$$[g(b) - g(a)] = [f(b) - f(a)]g'(c)$$



$g'(x) = 0$ for all x in (a, b) .



$g'(x) = f'(x)$ for all x in (a, b) .

$\lim_{x \rightarrow 0^+} x \log x$, has the following indeterminate form

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Select the correct option



$\infty \times \infty$.



$(0)(\infty)$.



$\frac{0}{0}$.



0^∞ .

If a function f is continuous on the closed interval then f attains its

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Select the correct option



extreme values in the closed interval.



extreme values at one point in the closed interval.



may or may not attain extreme values in the closed interval.



derivative in the closed interval.

The Mean Value Theorem says, suppose that f is differentiable on $[a, b]$, $f'(a) \neq f'(b)$, and μ is between $f'(a)$ and $f'(b)$.

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Select the correct option



Then $f'(c) \neq \mu$ for some c in (a, b) .



Then $f'(c) = \mu$ for some c in (a, b) .



Then $f'(c) = 0$ for some c in (a, b) .



Then $f'(c)$ may be 0 or μ for some c in (a, b) .

The function $f(x) = \begin{cases} x, & 0 \leq x < 1, \\ 2, & 1 \leq x \leq 2, \end{cases}$ is

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Select the correct option

☒	is nondecreasing on	$I = [0, 2]$.
☐	is nonincreasing on	$I = [0, 2]$.
☐	is having negative values on	$I = [0, 2]$.
☐	is undefined on	$I = [0, 2]$.

If $f : I \rightarrow \mathbb{R}$ has a derivative at $c \in I$, then f is

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Select the correct option

☒	continuous at c .
☐	discontinuous at c .
☐	undefined at c .
☐	having one sided limit.

If a function f is continuous on the closed interval then f attains its

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Select the correct option



extreme values in the closed interval.



extreme values at one point in the closed interval.



may or may not attain extreme values in the closed interval.



derivative in the closed interval.

The value of

$$\lim_{x \rightarrow +\infty} \frac{e^x}{x^2}$$

Select the correct option



0



$-\infty$.



∞ .



3

For the piecewise defined function $f(x) = \begin{cases} x^3, & x \leq 0, \\ x^2 \sin \frac{1}{x}, & x > 0, \end{cases}$ the value of $\lim_{x \rightarrow 0} f(x)$ is

Select the correct option

☒	0
☐	2
☐	1
☐	-1

Q. prove continuity for the given function. 2 questions

Q. second mean value theorem statement

Q using mean value theorem prove that $x^{-1}/x < \ln x < x^{-1}$ for $x > 1$

Q if $F'(x) = f(x)$ prove integral $f(x) = F(a) - F(b)$ for $[a, b]$

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Q if g is continuous at x_0 and f is continuous on $g(x_0)$ then show that $f \circ g$ is continuous at x_0 .

3.1 Monotone Sequences

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Areeb2113@gmail.com

3.1.1 Definition:

- (i) A sequence $\{a_n\}_{n \geq 1}$ said to be **monotonically increasing**, if $\{a_n\}_{n \geq 1} \geq a_n \forall n \geq 1$.
- (ii) A sequence $\{a_n\}_{n \geq 1}$ said to be **monotonically decreasing**, if $\{a_n\}_{n \geq 1} \leq a_n \forall n \geq 1$.

We can describe now the completeness property of the real numbers.

3.1.2 Completeness property

Every monotonically increasing sequence which is bounded above is convergent.

3.1.3 Theorem:

If $\{a_n\}_{n \geq 1}$ is monotonically decreasing and is bounded below, it is convergent.

Proof:

Follows from the following facts

- (i) $\{a_n\}_{n \geq 1}$ monotonically decreasing if and only if $\{a_n\}_{n \geq 1}$ monotonically increasing.
- (ii) $\{a_n\}_{n \geq 1}$ bounded below, if and only if $\{a_n\}_{n \geq 1}$ is bounded above.
- (iii) $\{a_n\}_{n \geq 1}$ is convergent if and only if $\{a_n\}_{n \geq 1}$ is convergent.

3.1.4 Examples:

- (i) Sequence $\{2n\}_{n \geq 1}$ is monotonically increasing and is not bounded above.
- (ii) Sequence $\{\frac{1}{n}\}_{n \geq 1}$ is monotonically decreasing and is not bounded below, say by 0.
- (iii) Sequence $\{(-1)^n\}_{n \geq 1}$ is neither monotonically increasing nor decreasing.

- (iv) Let $a_1 = 1$, $a_n = \frac{3a_{n-1} + 2}{6} \forall n \geq 2$. As shown in problem 2, module 0, $a_n \geq \frac{2}{3} \forall n$ and

Example: Find

$$\lim_{x \rightarrow \infty} \frac{2x^2 - x + 1}{3x^2 + 2x - 1}.$$

Example: Find

$$\lim_{x \rightarrow 1} \frac{x^8 - 1}{x^4 - 1}.$$

$$f(x) = \begin{cases} \frac{1}{x+2}, & x < -2, \\ x^2 - 5, & -2 < x \leq 3, \\ \sqrt{x+13}, & x > 3. \end{cases}$$

3.5 Continuity

Continuity (Informal):

- We say that f is *continuous at* x_0 if f is defined on an open interval (a, b) containing x_0 and

$$\lim_{x \rightarrow x_0} f(x) = f(x_0).$$

- We say that f is *continuous from the left at* x_0 if f is defined on an open interval (a, x_0) and

$$\lim_{x \rightarrow x_0^-} f(x) = f(x_0).$$

- We say that f is *continuous from the right at* x_0 if f is defined on an open interval (x_0, b) and

$$\lim_{x \rightarrow x_0^+} f(x) = f(x_0).$$

Continuity (Formal):

- A function f is continuous at x_0 if and only if f is defined on an open interval (a, b) containing x_0 and for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - f(x_0)| < \varepsilon \quad \text{whenever} \quad |x - x_0| < \delta.$$

- A function f is continuous from the right at x_0 if and only if f is defined on an interval $[x_0, b)$ and for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - f(x_0)| < \varepsilon \quad \text{whenever} \quad x_0 \leq x < x_0 + \delta.$$

Theorem: If f is integrable on $[a, b]$ and $[b, c]$, then f is integrable on $[a, c]$.

$$\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx. \quad (5.37)$$

Proof: See Lecture.

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5.4 Fundamental Theorem of Calculus

Theorem: If f is integrable on $[a, b]$ and $a \leq c \leq b$, then the function F defined by

$$F(x) = \int_c^x f(t) dt$$

satisfies a Lipschitz condition on $[a, b]$, and is therefore continuous on $[a, b]$.

Proof: If x and x' are in $[a, b]$, then

$$F(x) - F(x') = \int_c^x f(t) dt - \int_c^{x'} f(t) dt = \int_{x'}^x f(t) dt.$$

Since $|f(t)| \leq K$ ($a \leq t \leq b$) for some constant K ,

$$\left| \int_{x'}^x f(t) dt \right| \leq K|x - x'|, \quad a \leq x, x' \leq b.$$

Theorem: If f is integrable on $[a, b]$ and $a \leq c \leq b$, then the function F defined by

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$$F(x) = \int_c^x f(t) dt$$

satisfies a Lipschitz condition on $[a, b]$, and is therefore continuous on $[a, b]$.

Proof: Consequently,

$$|F(x) - F(x')| \leq K|x - x'|, \quad a \leq x, x' \leq b.$$

Theorem: If f is integrable on $[a, b]$ and $a \leq c \leq b$, then $F(x) = \int_c^x f(t) dt$ is differentiable at any point x_0 in (a, b) where f is continuous, with $F'(x_0) = f(x_0)$.

If f is continuous from the right at a , then $F'_+(a) = f(a)$. If f is continuous from the left at b , then $F'_-(b) = f(b)$.

Proof: Since

$$\frac{1}{x - x_0} \int_{x_0}^x f(x_0) dt = f(x_0).$$

We can write

$$\frac{F(x) - F(x_0)}{x - x_0} - f(x_0) = \frac{1}{x - x_0} \int_{x_0}^x [f(t) - f(x_0)] dt.$$

From this

$$\left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| \leq \frac{1}{|x - x_0|} \left| \int_{x_0}^x |f(t) - f(x_0)| dt \right|.$$

Since f is continuous at x_0 , there is for each $\varepsilon > 0$ a $\delta > 0$ such that

$$|f(t) - f(x_0)| < \varepsilon \quad \text{if} \quad |x - x_0| < \delta$$

and t is between x and x_0 .

Therefore, we have

First observe that if $L_2 \neq 0$, there is a $\delta_3 > 0$ such that

$$|g(x) - L_2| < \frac{|L_2|}{2}, \quad |g(x)| > \frac{|L_2|}{2} \quad \text{if } 0 < |x - x_0| < \delta_3.$$

Now suppose that $0 < |x - x_0| < \min(\delta_1, \delta_2, \delta_3)$. Then

$$\begin{aligned} \left| \left(\frac{f}{g} \right)(x) - \frac{L_1}{L_2} \right| &= \left| \frac{f(x)}{g(x)} - \frac{L_1}{L_2} \right| = \frac{|L_2 f(x) - L_1 g(x)|}{|g(x) L_2|} \\ &\leq \frac{2}{|L_2|^2} |L_2 f(x) - L_1 g(x)| \\ &= \frac{2}{|L_2|^2} |L_2 [f(x) - L_1] + L_1 [L_2 - g(x)]| \\ &\leq \frac{2}{|L_2|^2} [|L_2| |f(x) - L_1| + |L_1| |L_2 - g(x)|] \\ &\leq \frac{2}{|L_2|^2} (|L_2| + |L_1|) \varepsilon. \end{aligned}$$

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Example: Find

$$\lim_{x \rightarrow 2} \frac{9 - x^2}{x + 1} \quad \text{and} \quad \lim_{x \rightarrow 2} (9 - x^2)(x + 1).$$

Solution: If c is a constant, then $\lim_{x \rightarrow x_0} c = c$, and $\lim_{x \rightarrow x_0} x = x_0$. Therefore

$$\begin{aligned} \lim_{x \rightarrow 2} (9 - x^2) &= \lim_{x \rightarrow 2} 9 - \lim_{x \rightarrow 2} x^2 \\ &= \lim_{x \rightarrow 2} 9 - (\lim_{x \rightarrow 2} x)^2 \\ &= 9 - 2^2 = 5, \end{aligned}$$

and

$$\lim_{x \rightarrow 2} (x + 1) = \lim_{x \rightarrow 2} x + \lim_{x \rightarrow 2} 1 = 2 + 1 = 3.$$

Therefore,

$$\lim_{x \rightarrow 2} \frac{9 - x^2}{x + 1} = \frac{\lim_{x \rightarrow 2} (9 - x^2)}{\lim_{x \rightarrow 2} (x + 1)} = \frac{5}{3}$$

and

$$\lim_{x \rightarrow 2} (9 - x^2)(x + 1) = \lim_{x \rightarrow 2} (9 - x^2) \lim_{x \rightarrow 2} (x + 1) = 5 \cdot 3 = 15.$$

Example: The function

$$f(x) = 2x \sin \sqrt{x}$$

satisfies the inequality

$$|f(x)| < \varepsilon, \quad \text{whenever } 0 < x < \delta = \varepsilon/2.$$



3.3 Limits at $\pm\infty$

If f is defined on an interval (a, ∞) , then $f(x)$ approaches the limit L as x approaches ∞ , and write

$$\lim_{x \rightarrow \infty} f(x) = L.$$

For each $\varepsilon > 0$, there is a number β such that

$$|f(x) - L| < \varepsilon \quad \text{if } x > \beta.$$

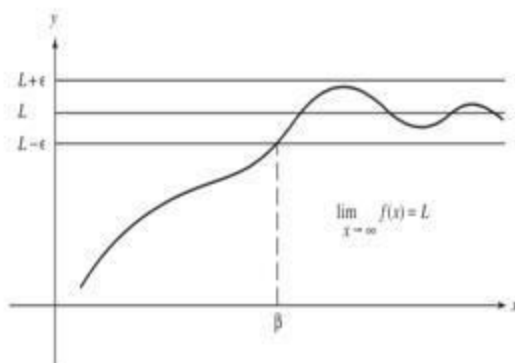


Figure 3.7: Limits at $\pm\infty$

Examples: Let

$$f(x) = 1 - \frac{1}{x^2}, \quad g(x) = \frac{2|x|}{1+x}, \quad \text{and} \quad h(x) = \sin x.$$

Find limits when $x \rightarrow \pm\infty$.

Solution: We have

$$\lim_{x \rightarrow \infty} f(x) = 1,$$

since

$$|f(x) - 1| = \frac{1}{x^2} < \varepsilon \quad \text{if } x > \frac{1}{\sqrt{\varepsilon}}.$$

We have

$$\lim_{x \rightarrow \infty} g(x) = 2,$$

since

$$|g(x) - 2| = \left| \frac{2x}{1+x} - 2 \right| = \frac{2}{1+x} < \frac{2}{x} < \varepsilon \quad \text{if } x > \frac{2}{\varepsilon}.$$

However, $\lim_{x \rightarrow \infty} h(x)$ does not exist, since h assumes all values between -1 and 1 in any semi-infinite interval (τ, ∞) . Discuss the limits at $x \rightarrow -\infty$.

$$f(x) = \frac{1}{x}, \quad g(x) = \frac{1}{x^2},$$

4.5 Rolle's Theorem

Theorem: Suppose that f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , and $f(a) = f(b)$. Then $f'(c) = 0$ for some c in the open interval (a, b) .

Proof: Since f is continuous on $[a, b]$, f attains a maximum and a minimum value on $[a, b]$. If these two extreme values are the same, then f is constant on (a, b) , so $f'(x) = 0$ for all x in (a, b) .

If the extreme values differ, then at least one must be attained at some point c in the open interval (a, b) , and $f'(c) = 0$.

4.6 The Mean Value Theorem

Theorem: Suppose that f is differentiable on $[a, b]$, $f'(a) \neq f'(b)$, and μ is between $f'(a)$ and $f'(b)$. Then $f'(c) = \mu$ for some c in (a, b) .

Proof: Suppose first that

$$f'(a) < \mu < f'(b)$$

and define

$$g(x) = f(x) - \mu x.$$

Then

$$g'(x) = f'(x) - \mu, \quad a \leq x \leq b,$$

and by our supposition $f'(a) < \mu < f'(b)$, we have

$$g'(a) < 0 \quad \text{and} \quad g'(b) > 0.$$

Since g is continuous on $[a, b]$, g attains a minimum at some point c in $[a, b]$.

Due to Lemma and $g'(a) < 0$ and $g'(b) > 0$, imply that there is a $\delta > 0$ such that

$$g(x) < g(a), \quad a < x < a + \delta, \quad \text{and} \quad g(x) < g(b), \quad b - \delta < x < b.$$

Therefore $c \neq a$ and $c \neq b$. Hence, $a < c < b$, and therefore $g'(c) = 0$, by the fact that if f is differentiable at a local extreme point x_0 then $f'(x_0) = 0$. From $g'(c) = 0$, $f'(c) = \mu$. The proof for the case where $f'(b) < \mu < f'(a)$ can be obtained by applying this result to $-f$.

Theorem: If f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , then

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

for some c in (a, b) .

4.11 Taylor's Theorem

Recall the following lemma: If f is differentiable at x_0 , then

$$f(x) = f(x_0) + [f'(x_0) + E(x)](x - x_0),$$

where E is defined on a neighborhood of x_0 and

$$\lim_{x \rightarrow x_0} E(x) = E(x_0) = 0.$$

To generalize this result, we first restate it: the polynomial

$$T_1(x) = f(x_0) + f'(x_0)(x - x_0),$$

which is of degree ≤ 1 and satisfies

$$T_1(x_0) = f(x_0), \quad T_1'(x_0) = f'(x_0).$$

$T_1(x)$ approximates f so well near x_0 that

$$\lim_{x \rightarrow x_0} \frac{f(x) - T_1(x)}{x - x_0} = 0.$$

Taylor polynomial: Suppose that f has n derivatives at x_0 and T_n is the polynomial of degree $\leq n$ such that

$$T_n^{(r)}(x_0) = f^{(r)}(x_0), \quad 0 \leq r \leq n.$$

How well does T_n approximate f near x_0 ?

Since T_n is a polynomial of degree $\leq n$, it can be written as

$$T_n(x) = a_0 + a_1(x - x_0) + \cdots + a_n(x - x_0)^n,$$

where $a_0, \dots, a_n$ are constants.

Differentiating yields

$$T_n^{(r)}(x_0) = r!a_r, \quad 0 \leq r \leq n,$$

We obtained a_r uniquely:

$$a_r = \frac{f^{(r)}(x_0)}{r!}, \quad 0 \leq r \leq n.$$

Therefore,

$$T_n(x) = f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \\ = \sum_{r=0}^n \frac{f^{(r)}(x_0)}{r!}(x - x_0)^r.$$

Special case of Taylor series: I
picked power series, I think it's right

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Areeb2113@gmail.com

Although each y_j occurs as a term of $\{x_n\}$, $\{y_j\}$ is not necessarily a subsequence of $\{x_n\}$, because if we write

$$y_j = x_{n_j}.$$

There is no reason to expect that $\{n_j\}$ is an increasing sequence as required in definition of subsequence.

However, it is always possible to pick a subsequence $\{n_{j_k}\}$ of $\{n_j\}$ that is increasing, and then the sequence $\{x_{n_{j_k}}\} = \{y_{j_k}\}$ is a subsequence of both $\{y_j\}$ and $\{x_n\}$.

2.6 Limit Superior and Limit Inferior

Theorem:

If $\{s_n\}$ is bounded above and does not diverge to $+\infty$, then there is a unique real number $\bar{s}$ such that, if $\varepsilon > 0$,

$$s_n < \bar{s} + \varepsilon \quad \text{for large } n \quad (2.4)$$

and

$$s_n > \bar{s} - \varepsilon \quad \text{for infinitely many } n. \quad (2.5)$$

- If $\{s_n\}$ is bounded below and does not diverge to $-\infty$, then there is a unique real number $\underline{s}$ such that, if $\varepsilon > 0$,

$$s_n > \underline{s} - \varepsilon \quad \text{for large } n \quad (2.6)$$

and

$$s_n < \underline{s} + \varepsilon \quad \text{for infinitely many } n. \quad (2.7)$$

Proof: Proof of the first part. Since $\{s_n\}$ is bounded above, there is a number β such that $s_n < \beta$ for all n . Since $\{s_n\}$ does not diverge to $+\infty$, there is a number α such that $s_n > \alpha$ for infinitely many n . If we define

$$M_k = \sup\{s_k, s_{k+1}, \dots, s_{k+r}, \dots\},$$

then $\alpha \leq M_k \leq \beta$, so $\{M_k\}$ is bounded. Since $\{M_k\}$ is nonincreasing (why?), it converges. Let

$$\bar{s} = \lim_{k \rightarrow \infty} M_k. \quad (2.8)$$

If $\varepsilon > 0$, then $M_k < \bar{s} + \varepsilon$ for large k , and since $s_n \leq M_k$ for $n \geq k$, $\bar{s}$ satisfies (2.4).

If (2.5) were false for some positive ε , there would be an integer K such that

$$s_n \leq \bar{s} - \varepsilon \quad \text{if } n \geq K.$$

However, this implies that

$$M_k \leq \bar{s} - \varepsilon \quad \text{if } k \geq K,$$

5.4. Fundamental Theorem of Calculus

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Example: If

$$f(x) = \begin{cases} x, & 0 \leq x \leq 1, \\ x + 1, & 1 < x \leq 2. \end{cases}$$

Then the function

$$F(x) = \int_0^x f(t) dt = \begin{cases} \frac{x^2}{2}, & 0 < x \leq 1, \\ \frac{x^2}{2} + x - 1, & 1 < x \leq 2, \end{cases}$$

is continuous on $[0, 2]$.

We can conclude

$$F'(x) = \begin{cases} x = f(x), & 0 < x < 1, \\ x + 1 = f(x), & 1 < x < 2. \end{cases}$$

$$F'_+(0) = \lim_{x \rightarrow 0+} \frac{F(x) - F(0)}{x} = \lim_{x \rightarrow 0+} \frac{(x^2/2) - 0}{x} = 0 = f(0),$$

$$F'_-(2) = \lim_{x \rightarrow 2-} \frac{F(x) - F(2)}{x - 2} = \lim_{x \rightarrow 2-} \frac{(x^2/2) + x - 1 - 3}{x - 2}$$

Date 10 Sep; 18

MTH 621

Sun Mon Tue Wed Thu Fri Sat

Long Questions:-

(Final 2nd)

- * Prove if f is integrable on $[a, b]$, then $|f|$ is integrable.
- * Is continuity implies differentiability? Explain with example.
- * If f is continuous on $[a, b]$, then it is bounded on $[a, b]$.
- * Find $\int_a^b f(x) dx$?
If $f(x) = \begin{cases} 0 & x \text{ is rational} \\ 1 & x \text{ is irrational} \end{cases}$
- * $h(x) = \sin \frac{1}{x}$ find derivative, by chain rule.
- * $g(x) = x^2$ show uniform continuity.
- * Define lower integral
- * Define GMVT.
- * Check $\lim_{x \rightarrow 2^-} f(x) = f(2)$
 $f(x) = \begin{cases} x^2 & 1 \leq x < 2 \\ n+1 & 1 < x \leq 2 \end{cases}$

MCQ's:-

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$$\rightarrow \lim_{n \rightarrow \infty} \frac{1}{n^p} = ?$$

$$\rightarrow \sin x \Big|_{-\pi/4}^{\pi/4} = ?$$

$$\rightarrow \text{lower sum ?}$$

$$\rightarrow \text{upper sum formula.}$$

$$\rightarrow \text{Comparison test}$$

$$\rightarrow \sum_{n=1}^{\infty} r^n = ?$$

$$\rightarrow \sup_{n \rightarrow \infty} S_n = \infty ?$$

$$\rightarrow e^x \rightarrow \text{in form of}$$

$$\int_a^b f(x) dx = ?$$

Good luck!
:)

Q. No. 1 Prove that if f is continuous on a finite closed Interval $[a, b]$, then f is bounded on $[a, b]$ (5)

Q. No. 2 Suppose that f is monotonic on (a, b) and define $\alpha = \inf f(x)$. If f is nondecreasing ^{as $x \rightarrow b$} then prove that $\lim_{x \rightarrow b} f(x) = \alpha$. (5)

Q. Define a continuously differentiable function.

Q. If Evaluate $\lim_{x \rightarrow \infty} \frac{\ln \sin x}{\ln x}$.

Q. Show that the function $g(x) = x^2$ is uniformly continuous on $[-x, x]$ for any finite x .

Q. State Generalize mean value theorem.

Example: If $s_n = [1 + (-1)^n]n$, then $\{s_n\}$ is bounded below ($s_n \geq 0$) but unbounded above.

$\{-s_n\}$ is bounded above ($-s_n \leq 0$) but unbounded below.

If $s_n = (-1)^n$, then $\{s_n\}$ is bounded.

If $s_n = (-1)^n n$, then $\{s_n\}$ is not bounded above or below.

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Theorem: A convergent sequence is bounded.

Proof: By taking $\varepsilon = 1$, we see that if $\lim_{n \rightarrow \infty} s_n = s$, then there is an integer N such that

$$|s_n - s| < 1 \quad \text{if } n \geq N.$$

Therefore,

$$|s_n| = |(s_n - s) + s| \leq |s_n - s| + |s| < 1 + |s| \quad \text{if } n \geq N,$$

and

$$|s_n| \leq \max\{|s_0|, |s_1|, \dots, |s_{N-1}|, 1 + |s|\}$$

for all n , so $\{s_n\}$ is bounded.

Theorem: A sequence $\{s_n\}$ converges to s if and only if every neighborhood of s contains s_n for all but finitely many n .

2.3 Monotonic Sequences

A sequence $\{s_n\}$ is *nondecreasing* if $s_n \geq s_{n-1}$ for all n , or *nonincreasing* if $s_n \leq s_{n-1}$ for all n .

A *monotonic sequence* is a sequence that is either nonincreasing or nondecreasing. If $s_n > s_{n-1}$ for all n , then $\{s_n\}$ is *increasing*, while if $s_n < s_{n-1}$ for all n , $\{s_n\}$ is *decreasing*.

Theorem:

- If $\{s_n\}$ is nondecreasing, then $\lim_{n \rightarrow \infty} s_n = \sup\{s_n\}$.
- If $\{s_n\}$ is nonincreasing, then $\lim_{n \rightarrow \infty} s_n = \inf\{s_n\}$.

Recall: The definition of supremum and infimum of a set.

Proof: Let $\beta = \sup\{s_n\}$. If $\beta < \infty$, then by definition if $\varepsilon > 0$ then

$$\beta - \varepsilon < s_N \leq \beta$$

for some integer N .

Since $s_N \leq s_n \leq \beta$ if $n \geq N$, it follows that

$$\beta - \varepsilon < s_n \leq \beta \quad \text{if } n \geq N.$$

Example: Prove that

$$\lim_{x \rightarrow 0^+} \sqrt{x} = 0.$$

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Solution: See Lecture.

Example: Show that

$$\lim_{x \rightarrow 0^+} \left(\frac{|x|}{x} + x \right) = 1, \quad \lim_{x \rightarrow 0^-} \left(\frac{|x|}{x} + x \right) = -1,$$

Solution: See Lecture

Theorem: A function f has a limit at x_0 if and only if it has left- and right-hand limits at x_0 , and they are equal. More specifically,

$$\lim_{x \rightarrow x_0} f(x) = L$$

if and only if

$$\lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) = L.$$

5 marks prove from this one

Theorem: If f is integrable on $[a, b]$ and $a \leq c \leq b$, then the function F defined by

$$F(x) = \int_c^x f(t) dt$$

satisfies a Lipschitz condition on $[a, b]$, and is therefore continuous on $[a, b]$.

Proof: Consequently,

$$|F(x) - F(x')| \leq K|x - x'|, \quad a \leq x, x' \leq b.$$

Theorem: If f is integrable on $[a, b]$ and $a \leq c \leq b$, then $F(x) = \int_c^x f(t) dt$ is differentiable at any point x_0 in (a, b) where f is continuous, with $F'(x_0) = f(x_0)$.

If f is continuous from the right at a , then $F'_+(a) = f(a)$. If f is continuous from the left at b , then $F'_-(b) = f(b)$.

Proof: Since

$$\frac{1}{x - x_0} \int_{x_0}^x f(t) dt = f(x_0).$$

We can write

$$\frac{F(x) - F(x_0)}{x - x_0} - f(x_0) = \frac{1}{x - x_0} \int_{x_0}^x [f(t) - f(x_0)] dt.$$

From this

$$\left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| \leq \frac{1}{|x - x_0|} \left| \int_{x_0}^x |f(t) - f(x_0)| dt \right|.$$

Since f is continuous at x_0 , there is for each $\varepsilon > 0$ a $\delta > 0$ such that

$$|f(t) - f(x_0)| < \varepsilon \quad \text{if} \quad |x - x_0| < \delta$$

and t is between x and x_0 .

Therefore, we have

$$\left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| < \varepsilon \frac{|x - x_0|}{|x - x_0|} = \varepsilon \quad \text{if} \quad 0 < |x - x_0| < \delta.$$

Hence, $F'(x_0) = f(x_0)$.

Theorem: If f is integrable on $[a, b]$ and $a \leq c \leq b$, then $F(x) = \int_c^x f(t) dt$ is differentiable at any point x_0 in (a, b) where f is continuous, with $F'(x_0) = f(x_0)$.

If f is continuous from the right at a , then $F'_+(a) = f(a)$. If f is continuous from the left at b , then $F'_-(b) = f(b)$.

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5- A theorem was from
Monotonic Function? 5 marks
6- evaluate $\lim_{x \rightarrow 0} \ln \sin x / \ln x$??
7- and Rest two was from the
last 5th chp of about integration
on closed interval. My Yesterday
paper;

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Mcqs were based on concepts
and 6,7 was from mcqs file of
quizz.

1- Define Removeable
continuity? 2 marks

2- Define chain rule for
composition of functions? 2
marks

3- Define Second mean theorem
of Integration?

4- Find uniform continuity of $2x$?

5- A theorem was from
Monotonic Function? 5 marks

6- evaluate $\lim_{x \rightarrow 0} \ln \sin x / \ln x$??

7- and Rest two was from the



Type a message

